How a Sidewalk Is Made and How It Works

01 What is a sidewalk made of?



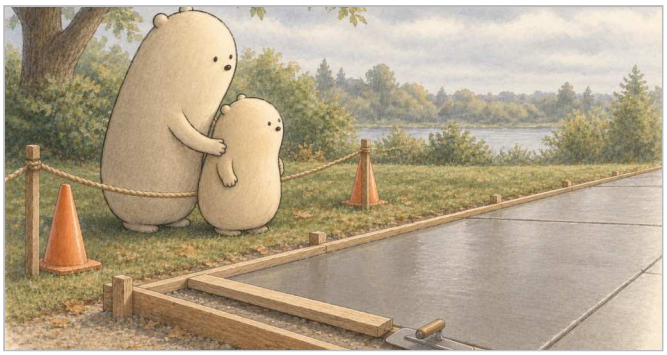
Concrete is gravel, sand, cement and water.

Look closer. Find the tiny stones and sandy specks on the chipped corner, then look at the four sealed jars beside it.

A little explanation. A sidewalk is made of concrete, and concrete is a mix of four things: gravel, sand, cement and water. Cement is a fine gray powder. It is only a small part of the mix, roughly one part in eight, and when it meets water it makes a paste that coats the sand and stones.

Try this. Look closely at a sidewalk square outside, or at a photo of one. Find tiny stones, speckles and a groove. Draw one square on paper and mark where the groove runs.

02 How does wet concrete turn hard?



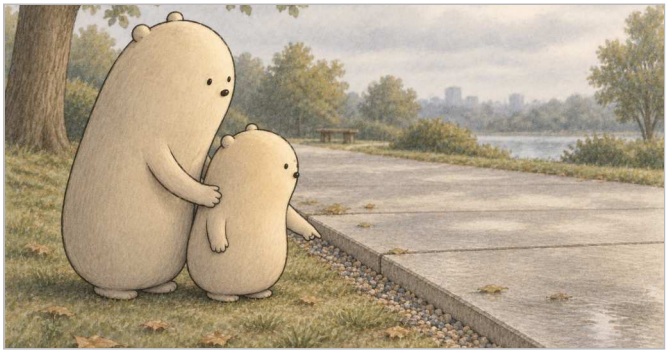
Wet concrete fills a wooden form and then hardens by a chemical reaction.

Look closer. Find the smooth wet slab inside its wooden frame, and the rope and cones that keep everyone back.

A little explanation. Wet concrete is poured into a wooden frame called a form, spread flat and smoothed. Then the cement and water begin a chemical reaction that links everything together. It hardens by that reaction, not just by drying out, which is why concrete can even harden under water.

Try this. Stir two spoonfuls of flour with a little water in a paper cup to make a thick paste. Stir in a handful of dry rice so the paste coats every grain. Leave the cup for a day, then tap it and feel how hard it got. Flour paste only dries out, but real concrete hardens by a chemical reaction.

03 Why is a sidewalk strong, and what holds it up?



Concrete is strong when squeezed, and packed gravel helps support the slab evenly.

Look closer. Look at the edge of the path, where a thin layer of packed gravel sits under the slab.

A little explanation. Concrete is very strong when it is squeezed, like when feet and bike wheels press down on it. It is weaker when it bends or stretches. A layer of packed gravel under the slab helps support it evenly and spread the weight out, so the slab bends less.

Try this. Stand a stick of sidewalk chalk on end and press down on top with one finger to see that it holds. Now lay the chalk across two books and press gently in the middle until it snaps. Chalk is a stand-in for concrete: strong when squeezed, weak when bent.

04 Why does a sidewalk have grooves?



A groove makes a planned weak line, so a crack tends to follow it.

Look closer. Follow the straight groove between two squares and find the thin crack that stays inside it.

A little explanation. A new slab shrinks a little as it hardens, and heat and cold make it grow and shrink all year. Concrete can crack when that happens. A groove makes the slab thinner along one line, so a crack tends to start there and run straight instead of wandering anywhere.

Try this. Look at the lines on a graham cracker. Break one cracker along a line. Then break another across the middle where there is no line. See which break is neat and straight.